* Naive Baye's Algorithm:

1. Probability
11. Baye's Theorem.

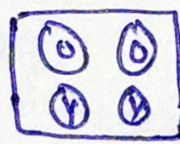
Independent Event

Rolling a Dice {1, 2, 3, 4, 5, 6}

$P(1) = \frac{1}{6}$ $P(2) = \frac{1}{6}$ $P(3) = \frac{1}{6}$

Dependent Events

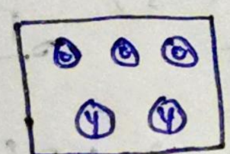
1. What is the probability of removing an orange marble and then a yellow marble.



$P(Y/O)$

$\to P(Y) = \frac{2}{4} = \frac{1}{2} \Rightarrow 2^{nd}$ Event

Removed the yellow marble



$\to P(O) = \frac{3}{5} \Rightarrow 1^{st}$ event

orange marble has been removed

$P(O \text{ and } Y) = \underset{1^{st}}{P(O)} * \underset{2^{nd}}{P(Y|O)}$

conditional prob.

$= \frac{3}{5} * \frac{1}{2} = \boxed{\frac{3}{10}}$

$$\boxed{P(A \text{ and } B) = P(A) * P(B|A)}$$

$$P(A \text{ and } B) = P(B \text{ and } A)$$

$$P(A) * P(B|A) = P(B) * P(A|B)$$

$$\boxed{P(A|B) = \frac{P(A) * P(B|A)}{P(B)}}$$ ⇒ Baye's Theorem

$P(A|B)$ = Probability of Event A given B has occured.

$P(A)$ = Probability of Event A

$P(B)$ = Probability of Event B

$\underbrace{x_1 \quad x_2 \quad x_3}_{\text{I/P Feature}}$			Dependent y
x_1	x_2	x_3	y
-	-	-	Yes
-	-	-	No
-	-	-	Yes
-	-	-	No

$$P(y/(x_1, x_2, x_3)) = \frac{P(y) * P((x_1, x_2, x_3)/y)}{P(x_1, x_2, x_3)}$$

$$= \frac{P(y) * P(x_1|y) * P(x_2|y) * P(x_3|y)}{P(x_1) * P(x_2) * P(x_3)}$$

New Test Data

$$P(Yes/(x_1, x_2, x_3)) = \frac{P(Yes) * P(x_1|Yes) * P(x_2/Yes) * P(x_3/Yes)}{\underbrace{\cancel{P(x_1) * P(x_2) * P(x_3)}}_{\text{constant}}} = \boxed{0.60} \Rightarrow Yes$$

$$P(No/(x_1, x_2, x_3)) = \frac{P(No) * P(x_1|No) * P(x_2/No) * P(x_3/No)}{\underbrace{\cancel{P(x_1) * P(x_2) * P(x_3)}}_{\text{constant}}} = \boxed{0.40} \Rightarrow No$$

Note: Remove or cut denominator because it is same for both Yes or No.

Problem Statement

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	week	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	week	Yes
4	Rain	Mild	High	week	Yes
5	Rain	Cold	Normal	week	Yes
6	Rain	Cold	Normal	Strong	No
7	Overcast	Cold	Normal	Strong	Yes
8	Sunny	Mild	High	week	No
9	Sunny	Cold	Normal	week	Yes
10	Rain	Mild	Normal	week	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	week	Yes
14	Rain	Mild	High	Strong	No

Outlook

	Yes	No	P(E/Yes)	P(E/No)
Sunny	2	3	2/9	3/5
Overcast	4	0	4/9	0/5
Rain	3	2	3/9	2/5

Temperature

	Yes	No	P(E/Yes)	P(E/No)
Hot	2	2	2/9	2/5
Mild	4	2	4/9	2/5
Cool	3	1	3/9	1/5

PLAY (Y/N)

		P(Yes)	P(No)
Yes	9	9/14	5/14
No	5		

→ Test (Sunny, Hot) → O/P

$$P(Yes/(Sunny, Hot)) = \frac{P(Yes) * P(Sunny|Yes) * P(Hot|Yes)}{\cancel{P(Sunny) * P(Hot)}}$$

$$= \frac{9}{\cancel{14}\,7} * \frac{2}{\cancel{9}} * \frac{2}{9} = \frac{2}{63} = 0.031$$

$$P(No/(Sunny, Hot)) = \frac{Pr(No) * P(Sunny|No) * P(Hot|No)}{\cancel{P(Sunny) * P(Hot)}}$$

$$= \frac{\cancel{5}}{\cancel{14}\,7} * \frac{3}{\cancel{5}} * \frac{2}{5} = \frac{3}{35} = 0.085$$

$$P(Yes/(Sunny, Hot)) = \frac{0.031}{(0.031+0.085)} = 0.27 = 27\%$$

$$P(No/(Sunny, Hot)) = \frac{0.085}{(0.031+0.085)} = 0.73 = 73\%$$

New Test

Outlook	Temperature	O/P
Sunny	Hot	73% => They will not play Tennis
		27% => They will play Tennis

Conclusion

o => Person is not going to play Tennis.